



Digital Design Verification

Lab Manual # 20 – RISC-V Datapath Implementation (Upper Immediate and Branches)

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Revision History

Revision Number	Revision Date	Revision By	Nature of Revision	Approved By
1.0	13/06/2024	Ali Aqdas	Complete manual	-
2.0	19/09/2025	Umer Farooq / Fawad Ahmed	Manual Update	



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Objective

The objective of this lab is to

- Extend the previously designed datapath to include branches and upper immediates

Tools

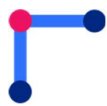
- Cadence Xcelium

Instructions

The students are required to design and submit the updated files for evaluation of this lab.

An additional file for branch comparison has to be added to the submission.

./<student-name>/	./rtl/	program_counter.sv inst_mem.sv data_mem.sv reg_file.sv imm_gen.sv alu_logic.sv top.sv
	./tb/	program_counter_tb.sv inst_mem_tb.sv data_mem_tb.sv reg_file_tb.sv imm_gen_tb.sv alu_logic_tb.sv top_tb.sv



RISC-V Datapath

The datapath designed so far that includes all the **R-Type**, **J-Type**, **S-Type**, and **I-Type Instructions**. The next goal is to extend the design to include upper immediate (U-Type) and Branches (B-Type).

Branches

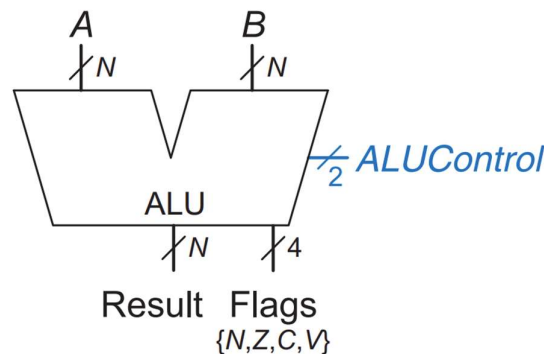
There are several types of branch instructions having B-Type Encoding and listed in the table below

imm _{12,10:5}	rs2	rs1	funct3	imm _{4:1,11}	op	B-Type
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op	funct3	funct7	Type	Instruction	Description	Operation
1100011 (99)	000	-	B	beq rs1, rs2, label	branch if =	if (rs1 == rs2) PC = BTA
1100011 (99)	001	-	B	bne rs1, rs2, label	branch if ≠	if (rs1 ≠ rs2) PC = BTA
1100011 (99)	100	-	B	blt rs1, rs2, label	branch if <	if (rs1 < rs2) PC = BTA
1100011 (99)	101	-	B	bge rs1, rs2, label	branch if ≥	if (rs1 ≥ rs2) PC = BTA
1100011 (99)	110	-	B	bltu rs1, rs2, label	branch if < unsigned	if (rs1 < rs2) PC = BTA
1100011 (99)	111	-	B	bgeu rs1, rs2, label	branch if ≥ unsigned	if (rs1 ≥ rs2) PC = BTA

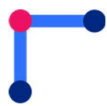
When the branch condition becomes true the processor jumps to the label listed in the instruction. Essentially, label is encoded as an immediate value that specifies the offset from current value of program counter. i.e. **PC = PC + Offset (Label)**.

It might seem evident, but for sake of clarity we explain that arithmetic logic unit (ALU) is busy computing program counter address, and we need to **we will use ALU for comparing the branch conditions**.



Here are the conditions of ALU for executing 6 branch instructions.

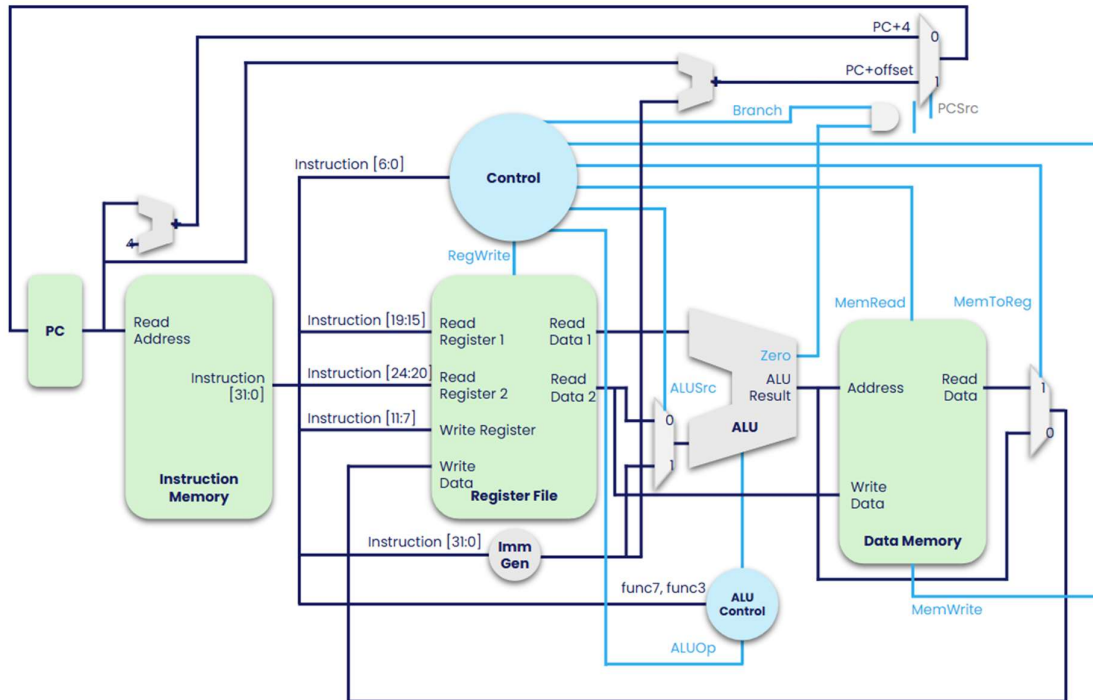
- If $RS1 - RS2 = 0$, this means that both values are equal, so we will set Zero Flag
Signal = 1 for beq otherwise Zero Flag Signal = 0 for bne
- If $RS1 - RS2 = -ve$, this means that $RS1 < RS2$, so we will set Negative Flag Signal = 1 for blt otherwise Negative Flag Signal = 0 for bgt. Similarly same condition for bltu & bgtu



Task 1

Extend the design of your datapath according to this diagram.

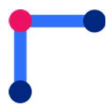
After extending your design, it may look along the lines of the following figure



Upper Immediate (U-Type)

The next goal is to extend the design to include upper immediate instructions, encoded as U-Type and listed in the table below.

imm _{31:12}				rd	op	U-Type
op	func3	func7	Type	Instruction	Description	Operation
0110111 (55)	-	-	U	lui rd, upimm	load upper immediate	rd = {upimm, 12'b0}
0010111 (23)	-	-	U	auipc rd, upimm	add upper immediate to PC	rd = {upimm, 12'b0} + PC



Task 2

Extend your datapath to include the upper immediate type instructions.

Now that your datapath is complete, it may look along the lines of following:

